Lab 7 — Fuzz Testing: Food Delivery

Step 4 — Run test_food_delivery.py (before fix)

`python -m pytest test_food_delivery.py`

```
> uv run food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza:2
Order: {'pizza': 2}

==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): 5.2
ETA in Minutes: 35

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): SAVE50
Discount: 50
```

before.png

```
============================= test session starts ==============================
platform linux -- Python 3.13.13, pytest-9.2.0, pluggy-1.7.0
rootdir: /mnt/data/programming/pesu/se/se-lab-pes1ug24am323/lab7
plugins: hypothesis-6.170.15
collected 3 items

test_food_delivery.py FFF                                                [100%]

=================================== FAILURES ===================================
__________________________ test_parse_order_no_crash ___________________________
  + Exception Group Traceback (most recent call last):
  |   File ".../test_food_delivery.py", line 8, in test_parse_order_no_crash
  |     def test_parse_order_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    |   File ".../food_delivery.py", line 5, in parse_order
    |     order[name.strip()] = int(qty)
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_parse_order_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    |   File ".../food_delivery.py", line 4, in parse_order
    |     name, qty = item.split(":")
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_parse_order_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    |   File ".../food_delivery.py", line 3, in parse_order
    |     for item in items_csv.split(","):
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_parse_order_no_crash(
    |     s=None,
    | )
    +------------------------------------
```

```
_____________________________ test_delivery_eta_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File ".../test_food_delivery.py", line 16, in test_delivery_eta_no_crash
  |     def test_delivery_eta_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    |   File ".../food_delivery.py", line 10, in delivery_eta
    |     return int(15 + float(distance_km) * 4)
    | OverflowError: cannot convert float infinity to integer
    | Failing test case: test_delivery_eta_no_crash(
    |     s='INFINITY',
    | )
    +---------------- 2 ----------------
    | ValueError: could not convert string to float: ''
    | Failing test case: test_delivery_eta_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | TypeError: float() argument must be a string or a real number, not 'NoneType'
    | Failing test case: test_delivery_eta_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________ test_coupon_discount_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File ".../test_food_delivery.py", line 24, in test_coupon_discount_no_crash
  |     def test_coupon_discount_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    |   File ".../food_delivery.py", line 14, in coupon_discount
    |     return int(code.upper().split("SAVE")[1])
    | IndexError: list index out of range
    | Failing test case: test_coupon_discount_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | AttributeError: 'NoneType' object has no attribute 'upper'
    | Failing test case: test_coupon_discount_no_crash(
    |     s=None,
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_food_delivery.py::test_parse_order_no_crash - ExceptionGroup: Hyp...
FAILED test_food_delivery.py::test_delivery_eta_no_crash - ExceptionGroup: Hy...
FAILED test_food_delivery.py::test_coupon_discount_no_crash - ExceptionGroup:...
============================== 3 failed in 0.66s ===============================
```

Step 6 — Re-run test_food_delivery.py (after fix)

```
python -m pytest test_food_delivery.py
```

```
> uv run food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza:1,burger:2
Order: {'pizza': 1, 'burger': 2}

==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): 5.2
ETA in Minutes: 35

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): SAVE50
Discount: 50
```

after.png

```
============================= test session starts ==============================
platform linux -- Python 3.13.13, pytest-9.2.0, pluggy-1.7.0
rootdir: /mnt/data/programming/pesu/se/se-lab-pes1ug24am323/lab7
plugins: hypothesis-6.170.15
collected 3 items

test_food_delivery.py ...                                                [100%]

============================== 3 passed in 0.45s ===============================
```